

JKBOSE 2026

CLASS 11TH

COMPLEX NUMBER

MATHS

LECTURE- 02

BY-MEHRAJ SIR



TOPICS TO BE COVERED

1

Algebra of Complex Numbers (Addition, Multiplication, Subtraction, Division)

2

Modulus and Conjugate of a Complex Number

3

Related Problems

* $\text{Re}(z) = 0 \rightarrow z$ is purely imaginary
 * $\text{Im}(z) = 0 \rightarrow z$ is purely real

$$z = a + ib, \quad a, b \in \mathbb{R}.$$

$$i = j = \sqrt{-1}$$

* $a = \text{Re}(z) \rightarrow$ Real part of z
 * $b = \text{Im}(z) \rightarrow$ Imaginary part of z .

EXAMPLE

If $4x + i(3x - y) = 3 + i(-6)$, where x and y are real numbers, then find the values of x and y .

(Sol) Given that $4x + i(3x - y) = 3 + i(-6)$

So, $4x = 3$ and $3x - y = -6$.

$\Rightarrow x = \frac{3}{4}$ & $y = 3x + 6$.

$\Rightarrow x = \frac{3}{4}$, $y = 3 \times \frac{3}{4} + 6 = \frac{9 + 24}{4} = \frac{33}{4}$

$\Rightarrow x = \frac{3}{4}, y = \frac{33}{4}$

Hence the result.

* $z_1 = z_2$
 $\text{Re}(z_1) = \text{Re}(z_2)$
 $\text{Im}(z_1) = \text{Im}(z_2)$

QUESTION

If $(a + 2b) + i(a - 2b) = 3 + 7i$, then the value of a and b is:

1 ~~$a = 1, b = 0$~~

2 $a = 5, b = -1$

3 ~~$a = 1, b = 6$~~

4 ~~$a = -1, b = 5$~~

(Sol) Given that $a + 2b + i(a - 2b) = 3 + 7i$

So, $a + 2b = 3$ — (1)

& $a - 2b = 7$ — (2)

Adding (1) & (2), we get;

~~$a + 2b + a - 2b = 3 + 7$~~

$\Rightarrow 2a = 10$

$\Rightarrow a = 5$

So, $5 - 2b = 7$

$\Rightarrow 2b = 5 - 7$

$\Rightarrow 2b = -2$

$\Rightarrow b = -1$

So, $a = 5, b = -1$

EXAMPLE

$$(a+b)(a-b) = a^2 - b^2.$$

$i^2 = -1.$

$$-9i^2 = -9 \times -1 = 9.$$



Express the following complex numbers in $a+ib$ form :

i. $\frac{1}{3-2i}$ ii) $\frac{2}{6+3i}$

ii. (i) (Sol) let $z = \frac{1}{3-2i} = \frac{1}{3-2i} \times \frac{3+2i}{3+2i}$

$$\Rightarrow z = \frac{3+2i}{(3-2i)(3+2i)} = \frac{3+2i}{3^2 - (2i)^2}$$
$$\Rightarrow z = \frac{3+2i}{9-4i^2} = \frac{3+2i}{9+4} = \frac{3+2i}{13}$$

$$\Rightarrow z = \left(\frac{3}{13}\right) + \left(\frac{2}{13}\right)i$$

(ii) (Sol) let $z = \frac{2}{6+3i} = \frac{2}{6+3i} \times \frac{6-3i}{6-3i}$

$$\Rightarrow z = \frac{2(6-3i)}{(6+3i)(6-3i)} = \frac{12-6i}{6^2 - (3i)^2}$$

$$\Rightarrow z = \frac{12-6i}{36-9i^2} = \frac{12-6i}{36+9} = \frac{12-6i}{45}$$

$$\Rightarrow z = \left(\frac{12}{45}\right) + \left(\frac{-6}{45}\right)i$$

$$\Rightarrow z = \left(\frac{4}{15}\right) + \left(\frac{-2}{15}\right)i$$

QUESTION

$$(a-b)(a+b) = a^2 - b^2$$



The real part of the complex number $\frac{13}{3+2i}$ is :

☒ 13

☒ 3

☐ 2

☐ 5

$$\text{Sol) let } z = \frac{13}{3+2i} = \frac{13}{3+2i} \times \frac{(3-2i)}{(3-2i)}$$

$$\Rightarrow z = \frac{13(3-2i)}{(3+2i)(3-2i)} = \frac{39-26i}{3^2-(2i)^2}$$

$$\Rightarrow z = \frac{39-26i}{9-4i^2} = \frac{39-26i}{9+4}$$

$$\Rightarrow z = \left(\frac{39}{13}\right) + \left(\frac{-26}{13}\right)i$$

$$\Rightarrow z = 3 - 2i$$

$$\Rightarrow \text{Re}(z) = 3$$

$$\& \text{Im}(z) = -2$$

QUESTION

$$3^2 - (2i)^2 = 9 - 4i^2 \\ = 9 + 4 = 13$$



The imaginary part of the complex number $\frac{26}{3+2i}$ is :

1 $13 \times$

2 $-3 \times$

3 $-4 \checkmark$

4 $4 \times$

$$\text{Sol) let } z = \frac{26}{3+2i} = \frac{26}{3+2i} \times \frac{3-2i}{3-2i}$$

$$\Rightarrow z = \frac{26(3-2i)}{(3+2i)(3-2i)} = \frac{26(3-2i)}{9+4}$$

$$\Rightarrow z = \frac{26(3-2i)}{13} = 6-4i$$

$$\Rightarrow z = 6-4i$$

$$\text{So } \operatorname{Re}(z) = 6$$

$$\& \operatorname{Im}(z) = -4$$

EXAMPLE

Find the values of θ if $\frac{3+2i\sin\theta}{1-2i\sin\theta}$ is Purely real or Purely imaginary.

$$(Sol) \text{ let } z = \frac{3+2i\sin\theta}{1-2i\sin\theta}$$

$$\Rightarrow z = \frac{(3+2i\sin\theta)}{(1-2i\sin\theta)} \times \frac{(1+2i\sin\theta)}{(1+2i\sin\theta)}$$

$$\Rightarrow z = \frac{(3+2i\sin\theta)(1+2i\sin\theta)}{(1-2i\sin\theta)(1+2i\sin\theta)}$$

$$\Rightarrow z = \frac{3 + 6i\sin\theta + 2i\sin\theta + 4i^2\sin^2\theta}{1^2 - (2i\sin\theta)^2}$$

$$\Rightarrow z = \frac{3 + 8i\sin\theta - 4\sin^2\theta}{1 - 4i^2\sin^2\theta}$$

$$\Rightarrow z = \frac{(3 - 4\sin^2\theta) + i(8\sin\theta)}{(1 + 4\sin^2\theta)}$$

$$\Rightarrow z = \left(\frac{3 - 4\sin^2\theta}{1 + 4\sin^2\theta} \right) + i \left(\frac{8\sin\theta}{1 + 4\sin^2\theta} \right)$$

Case 1st: If z is purely real, then

$$\operatorname{Im}(z) = 0$$

$$\Rightarrow \frac{8 \sin \theta}{1 + 4 \sin^2 \theta} = 0$$

$$\Rightarrow 8 \sin \theta = 0$$

$$\Rightarrow \sin \theta = 0$$

$$\Rightarrow \boxed{\theta = n\pi, n \in \mathbb{Z}}$$

$$\theta = 0, \pi, 2\pi, 3\pi, 4\pi, \dots \\ -\pi, -2\pi, -3\pi, \dots$$

$$\sin\left(\pi - \frac{\pi}{3}\right) = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

$$\sin\left(\pi + \frac{\pi}{3}\right) = -\sin\left(\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{2}$$

Case 2nd: If z is purely imaginary, then

$$\operatorname{Re}(z) = 0$$

$$\Rightarrow \frac{3 - 4 \sin^2 \theta}{1 + 4 \sin^2 \theta} = 0$$

$$\Rightarrow 3 - 4 \sin^2 \theta = 0$$

$$\Rightarrow 4 \sin^2 \theta = 3$$

$$\Rightarrow \sin^2 \theta = \frac{3}{4}$$

$$\Rightarrow \boxed{\sin \theta = \pm \frac{\sqrt{3}}{2}}$$

$$\Rightarrow \theta = \frac{\pi}{3}, \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}, 2\pi - \frac{\pi}{3}, \\ 2\pi + \frac{\pi}{3}, 3\pi - \frac{\pi}{3}, 3\pi + \frac{\pi}{3}, \dots$$

$$\Rightarrow \boxed{\theta = n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}}$$

$$* \sin\left(2\pi + \frac{\pi}{3}\right) = \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}.$$

$$* \sin\left(2\pi - \frac{\pi}{3}\right) = -\sin\left(\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{2}.$$

QUESTION (NCERT EXAMPLER)

$$7 = 2(3) + 1, 2(4) - 1.$$



The real value of θ for which the expression $\frac{1+i\cos\theta}{1-2i\cos\theta}$ is a real number is :

~~1~~ $n\pi + \frac{\pi}{4}$

~~2~~ $n\pi + (-1)^n \frac{\pi}{4}$

~~3~~ $2n\pi \pm \frac{\pi}{2} = \frac{4n\pi \pm \pi}{2}$

~~4~~ None of these

$(4n \pm 1) \frac{\pi}{2}$

(Sol) let $z = \frac{1+i\cos\theta}{1-2i\cos\theta}$

$$\Rightarrow z = \frac{(1+i\cos\theta)(1+2i\cos\theta)}{(1-2i\cos\theta)(1+2i\cos\theta)}$$

$$\Rightarrow z = \frac{1+2i\cos\theta+i\cos\theta+2i^2\cos^2\theta}{1-4i^2\cos^2\theta}$$

$$\Rightarrow z = \frac{(1-2\cos^2\theta) + i(3\cos\theta)}{(1+4\cos^2\theta)}$$

$$\Rightarrow z = \left(\frac{1-2\cos^2\theta}{1+4\cos^2\theta} \right) + i \left(\frac{3\cos\theta}{1+4\cos^2\theta} \right)$$

If z is purely real, then
 $\text{Im}(z) = 0$

$$\frac{3\cos\theta}{1+4\cos^2\theta} = 0$$

$$\Rightarrow 3\cos\theta = 0$$

$$\Rightarrow \cos\theta = 0$$

$$\theta = (2n+1) \frac{\pi}{2}$$

EXAMPLE

$$\begin{aligned} & \overbrace{2+3i+4+6i} \\ &= 6+9i \end{aligned}$$

Express each of the following in the form $a+ib$:

I. $(3 + i\sqrt{5})(4 + i\sqrt{5})$

II. $6(1 + 2i) + 10(2 + 5i)$ *Homework*

III. $\frac{3+2i}{3-2i}$

#

The Modulus and the Conjugate of a Complex Number

Modulus of a Complex number:

let $z = a+ib$ be any Complex number,
then the modulus of z is denoted & defined as,

$$|z| = \sqrt{a^2 + b^2} = \sqrt{(\operatorname{Re}(z))^2 + (\operatorname{Im}(z))^2}.$$

$$* \quad z = 2-3i$$

$$\Rightarrow |z| = \sqrt{(2)^2 + (-3)^2}$$

$$\Rightarrow |z| = \sqrt{4+9}$$

$$\Rightarrow |z| = \sqrt{13}$$

Conjugate of a Complex number:

let $z = a+ib$ be any Complex number
then the Conjugate of z is denoted & defined as:

$$\overline{z} = a-ib$$

$$* \quad z = 2-3i \Rightarrow \overline{z} = 2+3i.$$

$$* \quad z\overline{z} = (a+ib)(a-ib) = a^2 - (ib)^2.$$

$$\Rightarrow z\overline{z} = a^2 - i^2b^2$$

$$\Rightarrow z\overline{z} = a^2 + b^2.$$

$$\Rightarrow z\overline{z} = \left(\sqrt{a^2 + b^2}\right)^2$$

$$\Rightarrow z\overline{z} = |z|^2.$$

QUESTION

The modulus of a Complex number $1 + i$ is

1

$-\sqrt{2}$ ✗ $z = 1 + i$

$\Rightarrow |z| = \sqrt{(1)^2 + (1)^2}$

2

2 ✗

$\Rightarrow |z| = \sqrt{2}$

3

$\sqrt{2}$ ✓✓

4

-2 ✗

QUESTION

If $a + ib = c + id$, then

$$a = c, b = d.$$

1 ☒ $a^2 + c^2 = 0$

2 ☒ $b^2 + c^2 = 0$

3 ☒ $b^2 + d^2 = 0$

4 ☒ $a^2 + b^2 = c^2 + d^2$

$$(su) |a+ib| = |c+id|$$

$$\Rightarrow \sqrt{a^2+b^2} = \sqrt{c^2+d^2}$$

Sq. on b/s, we get:

~~$$\sqrt{a^2+b^2} = \sqrt{c^2+d^2}$$~~

$$\Rightarrow a^2 + b^2 = c^2 + d^2$$

QUESTION

The Conjugate of the complex number $3 + 3i$ is :

- 1** ✓✓ $3 - 3i$ ✓
- 2** ✗ $-3 + 3i$ ✗
- 3** ✗ $-3 - 3i$ ✗
- 4** ✗ None of the above

Theorem

$$z = 3 + 0 \cdot i$$

$$\bar{z} = 3 - 0 \cdot i = 3 = z$$

If Z, Z_1 and Z_2 are any complex numbers , then

$$(i) \quad \overline{(\bar{Z})} = Z$$

$$(ii) \quad Z\bar{Z} = |Z|^2$$

$$(iii) \quad Z + \bar{Z} = 2\text{Re}(Z)$$

$$(iv) \quad Z - \bar{Z} = 2i \text{Im}(Z)$$

$$(v) \quad Z = \bar{Z} \text{ iff } Z \text{ is purely real}$$

$$(vi) \quad Z + \bar{Z} = 0 \text{ iff } Z \text{ is purely imaginary}$$

$$(vii) \quad \overline{Z_1 + Z_2} = \bar{Z}_1 + \bar{Z}_2$$

$$(viii) \quad \overline{Z_1 - Z_2} = \bar{Z}_1 - \bar{Z}_2$$

$$(ix) \quad \overline{Z_1 Z_2} = \bar{Z}_1 \bar{Z}_2$$

$$(x) \quad \overline{\left(\frac{Z_1}{Z_2}\right)} = \frac{\bar{Z}_1}{\bar{Z}_2}, Z_2 \neq 0$$

QUESTION

$$z\bar{z} = |z|^2.$$

The value of $(z + 3)(\bar{z} + 3)$ is equivalent to

1

$$|z + 3|^2 \checkmark$$

2

$$|z - 3| \times$$

3

$$z^2 + 3 \times$$

4

None of these $\times$

$$\begin{aligned} (z+3)(\bar{z}+3) \\ &= (z+3)(\overline{\bar{z}+3}) \\ &= (z+3)(\overline{z+3}) \\ &= |z+3|^2 \end{aligned}$$

QUESTION



If $\frac{1 - ix}{1 + ix} = a + ib$, then $a^2 + b^2 =$

Homework
Problem.
#.

1

1

2

0

3

-1

4

None of these

QUESTION

If $\left| \frac{z-2}{z-4} \right| = 1$, then $\text{Re}(z)$ is equal to :

Homework.

1 3

2 0

3 -3

4 None of these

$$(2+3i) - (4+6i)$$
$$2-4+3i-6i$$
$$= -2-3i$$

$$z = -2+4i$$

$$\bar{z} = -2-4i$$

Thank You

Jan_Mehra1729. ✓✓

